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**Campus Event Management System (CEMS) –
Project Plan**

Metropolia University of Applied
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Software Engineering Project 1

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1 Project overview

CEMS is an event management system that will be designed to assist students and teachers in navigating events happening across the campus. With vast amounts of events taking place, it becomes difficult for users to search for events information and organizers often struggle to manage RSVPs and track attendance closely. This project aims to solve that problem with a centralized functional app where students and teachers can browse events, view details, receive notifications, and RSVP to confirm their participation. Event organizers and administrators can create events and monitor attendance.

2 Project objectives

- Enable admin and event organizers to add event, event date and time, and location within 5 minutes.
- Display all upcoming events in a scrollable list
- Enable users to search for events by title or filter by tags within 1.5 minutes
- Enable users to RSVP for events and add specific details to the host/organizers asked for in less than 5 minutes.
- Enable users to authenticate to the app to save events the user had signed up to and to save personal details to fill out future forms.

- Enable admin and event organizers to track attendance of events they have created.
- Implement JWT for secure user authentication.
- Frontend must be responsive, meaning that the UI should function on all screens and aspect ratios.

3 Scope and deliverables

The scope of this project Campus Event Management System is defined by the functional and technical requirements established in the first sprint of the DevOps lifecycle. This project focuses on the transition from a Java-based desktop application to a fully automated containerized environment.

3.1 Project Scope

- **Functional Scope:** This system support two primary user roles: Students as: viewer or attendees and Teachers or Admin as: organizers. The key functionalities include user authentication, a centralized event dashboard, RSVP submission, and attendance tracking.
- **Technical Scope:** Technical Scope: This project implementation is restricted to a three-tier architecture using Java (JavaFX UI), MariaDB for relational data management, and JDBC for database connectivity.
- **DevOps Scope:** This project scope extends beyond code to include the implementation of a CI/CD pipelining using Jenkins, automated unit testing with JUnit, and application packaging via Docker.

3.2 Project Deliverables

- **Software Artifacts:** The fully functional JavaFX application integrated with a MariaDB database.
- **DevOps Artifact:** It is a configured Jenkins file for automated build or test pipelines and a docker file for containerized deployment.
- **Quality Artifacts:** The comprehensive suite of JUnit tests and code coverage reports.
- **Documentation:** A well-maintained GitHub repository featuring a clear commit history, README, and the final project report including UML and ER diagrams.

4 Project timeline

Sprint	Start Date	End Date	Aims
Sprint 1	13-01-2026	27-01-2026	• Create a product vision and project plan. • Select Technologies. • Set Up Project Management Tools (Trello) • Create the Initial Product Backlog
Sprint 2	27-01-2026	10-02-2026	• Design basic UI on Figma. • Integrate database and implement basic. • Initial features such as event listing. • Unit testing for basic checking.
Sprint 3	10-02-2026	03-03-2026	• Expansion with further detailed features. • User Authentication. • Integration of Jenkins and Docker.
Sprint 4	03-03-2026	17-03-2026	• Bug fixing and UI polishing. • Complete documentation and finalize the project for submission.

5 Resource allocation

- Team members & Roles:
 - Aroush Irfan: Scrum master for sprint 2
 - Full stack Developer (Focusing on Backend and Database for sprint 1 and 2, Frontend for sprint 3 and 4)
 - Software Testing Engineer
 - Jiya Jameela Kanhiranthan Poyil: Scrum master for sprint 3,
 - Full stack Developer (Frontend Developer and UI/UX Design for sprint 1 and 2, Backend and Database for sprint 3 and 4)
 - Software Testing Engineer
 - Puntawat Subhamani: Scrum master for sprint 1,
 - Full stack Developer (Focusing on Backend and Database for sprint 1 and 2, Frontend for sprint 3 and 4)
 - Software Debugger
 - Sailesh Karki: Scrum master for sprint 4,
 - Full stack Developer (Focusing on Frontend and UI/UX Design for sprint 1 and 2, Backend and Database for sprint 3 and 4)
 - Software Debugger
- Software, Hardware, and Tools:
 - Programming Language: Java (JavaFX Frontend, JDBC), SQL (MariaDB)
 - Software: IntelliJ IDEA

- Hardware: Computer running Windows, macOS, or Linux
- Testing: JUnit

6 Risk management

The team has adopted a proactive risk management framework to identify and address potential bottlenecks within the software development lifecycle. Each risk is evaluated based on its Likelihood and Impact to ensure the stability of the delivery pipeline.

6.1 Technical and Infrastructure Risks

- Risk: Environment Inconsistency
 - Description: Differences in local Java or Maven configurations among the four members causing build failures.
 - Likelihood: Medium
 - Impact: High
 - Mitigation: The team will utilize a standard pom.xml file and leverage docker to ensure environmental parity.
- Risk: Database Connectivity Failures
 - Description: Failure to establish a stable JDBC connection between the Java application and MariaDB, particularly within a Docker network.
 - Likelihood: Medium
 - Impact: High

- Mitigation: The team will implement environment-based configuration files and conduct early integration testing in Sprint 2.

6.2 Process and Pipeline Risks

- Risk: CI/CD Pipeline Configuration Errors
 - Description: Technical hurdles in configuring Jenkins triggers or automated test execution scripts.
 - Likelihood: High
 - Impact: Medium
 - Mitigation: The team will strictly follow course demonstrations and use the mandatory temperature converter application as a baseline for pipeline configuration.
- Risk: Scope Creep and Time management
 - Description: Attempting to implement non-essential features that delay the delivery of mandatory DevOps components.
 - Likelihood: Low
 - Impact: Medium
 - Mitigation: The Scrum Master will ensure strict adherence to the Product Backlog and prioritize tasks in Trello during each sprint Planning meeting.

Testing and quality assurance

- Types of testing: Unit testing
- Criteria for success:

- Finish all the task required in project objectives without any critical errors within a given amount of time.
- Tools: JUnit

7 Documentation and reporting

Throughout the project, we plan to provide a user manual to support both users and developers in navigating the application and understanding the objectives of the project. Our group will maintain consistent reporting through daily scrums, internal meetings, and task reviews to keep track of the overall progress. Trello will be used as the primary platform for organizing and tracking the tasks. In addition to internal reporting, the sprint review reports will be submitted to showcase our progress to the instructor through each sprint.